

COMMUNICATION ENGINEERING

MOD-4

Q1. 1. Time division multiplexing with diagram and its Application.

This happens when data transmission rate of media is greater than that of the source, and each signal is allotted a definite amount of time. These slots are so small that all transmissions appear to be parallel. In frequency division multiplexing all the signals operate at the same time with different frequencies, but in time division multiplexing all the signals operate with same frequency at different times.

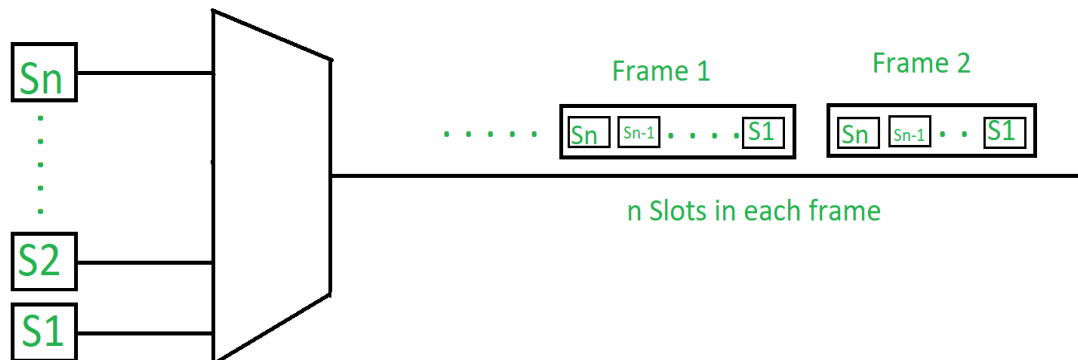
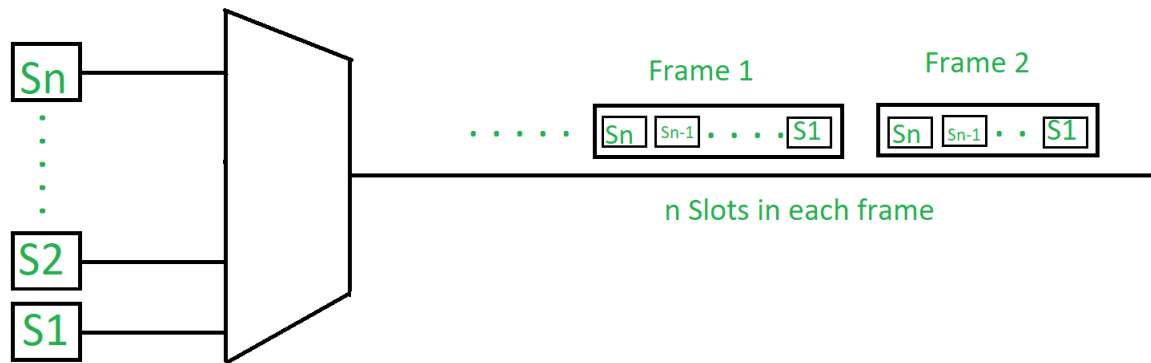


Fig Time Division Multiplexing

Synchronous TDM

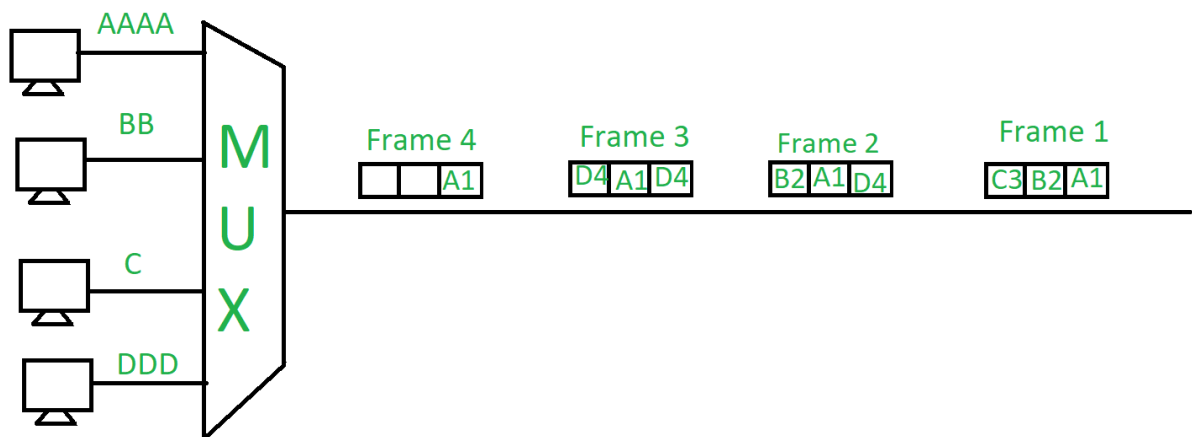
The time slots are pre-assigned and fixed. This slot is even given if the source is not ready with data at this time. In this case the slot is transmitted empty. It is used for multiplexing digitized voice stream.



Synchronous Time Division Multiplexing

Asynchronous (or statistical) TDM

The slots are allocated dynamically depending on the speed of source or their ready state. It dynamically allocates the time slots according to different input channel's needs, thus saving the channel capacity.



Asynchronous (or statistical) Multiplexing

Advantages:-

- It can deliver fast data;
- It can protect data from degradation;
- It can utilize expensive transmission lines better than most other methods;
- It works well with satellite communications.

- its ability to increase bandwidth. It does so by enabling several signals to be transmitted

Disadvantages:-

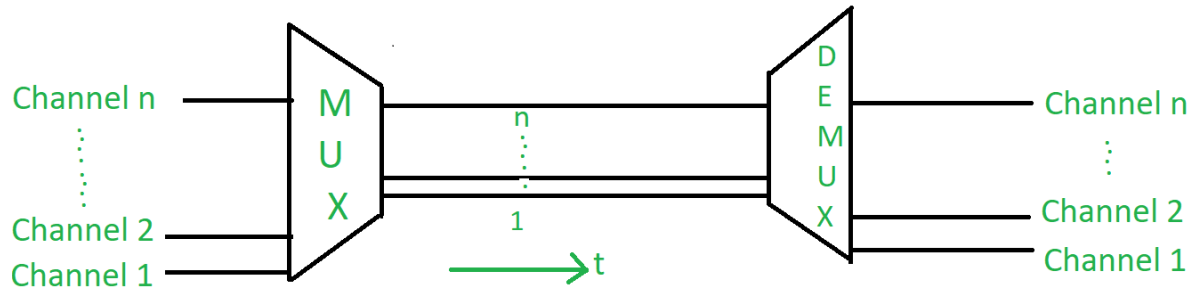
- A failure in one channel can cause the entire system to fail
- It's more complex than other methods, so there's more room for errors
- It is more vulnerable to noise than other forms of multiplexing. While baseband signals can travel long distances without encountering noise, bandpass signals are more prone to experiencing interference as they pass through a medium.
- It requires a complex and expensive regenerative repeater, which is used to regenerate the signal at each repeater.

Applications:-

A common application of time-division multiplexing is in digital telephony. There are two main types of telephone lines: analog and digital. Analog lines use frequency-division multiplexing, where each signal is sent on its own frequency band. Digital lines use time-division multiplexing, where the signals are interleaved at different times to prevent them from interfering with each other.

Q2. FDM with diagram and its Application.

In this, a number of signals are transmitted at the same time, and each source transfers its signals in the allotted frequency range. There is a suitable frequency gap between the 2 adjacent signals to avoid over-lapping. Since the signals are transmitted in the allotted frequencies so this decreases the probability of collision. The frequency spectrum is divided into several logical channels, in which every user feels that they possess a particular bandwidth. A number of signals are sent simultaneously at the same time allocating separate frequency bands or channels to each signal. It is used in radio and TV transmission. Therefore to avoid interference between two successive channels Guard bands are used.



Advantages of Frequency Division Multiplexing (FDM):

Efficient Use of Bandwidth: FDM allows multiple signals to be transmitted over a single communication channel, which can lead to more efficient use of available bandwidth.

No Time Synchronization Required: FDM does not require precise time synchronization between the transmitting and receiving devices, making it easier to implement.

Low Implementation Cost: FDM is a relatively simple technique that does not require sophisticated hardware or software, making it less expensive to implement.

Disadvantages of Frequency Division Multiplexing (FDM):

Limited Capacity: FDM is limited in terms of the number of signals that can be transmitted over a single communication channel, which can be a disadvantage in applications where a large number of signals need to be transmitted.

Interference: FDM can be susceptible to interference from other signals transmitted on nearby frequencies, which can degrade the quality of the transmitted signals.

Difficulty in Assigning Frequencies: FDM requires careful assignment of frequencies to different signals to avoid interference, which can be a complex and time-consuming process.

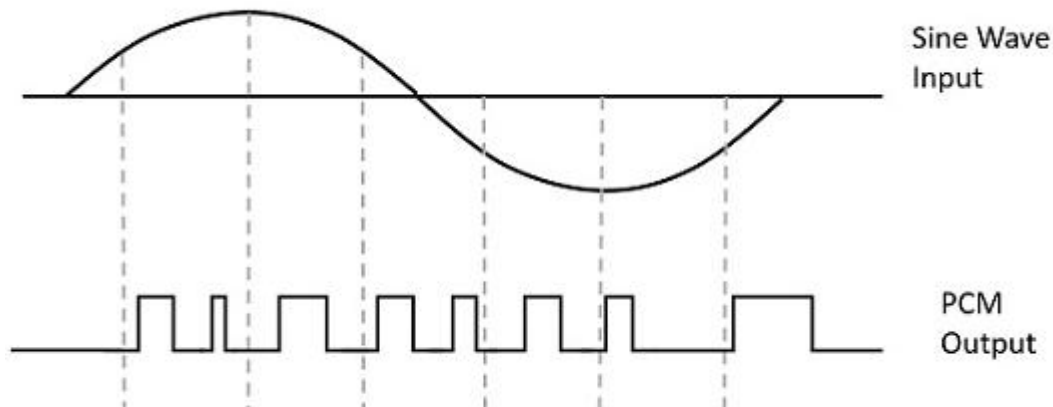
Application of FDM:

1. In the first generation of mobile phones, FDM was used.
2. The use of FDM in television broadcasting
3. FDM is used to broadcast FM and AM radio frequencies.

Q5. Explain Pulse code modulation and delta modulation.

A signal is Pulse Code modulated to convert its analog information into a binary sequence, i.e., 1s and 0s. The output of a Pulse Code Modulation (PCM) will resemble a binary sequence.

The following figure shows an example of PCM output with respect to instantaneous values of a given sine wave.



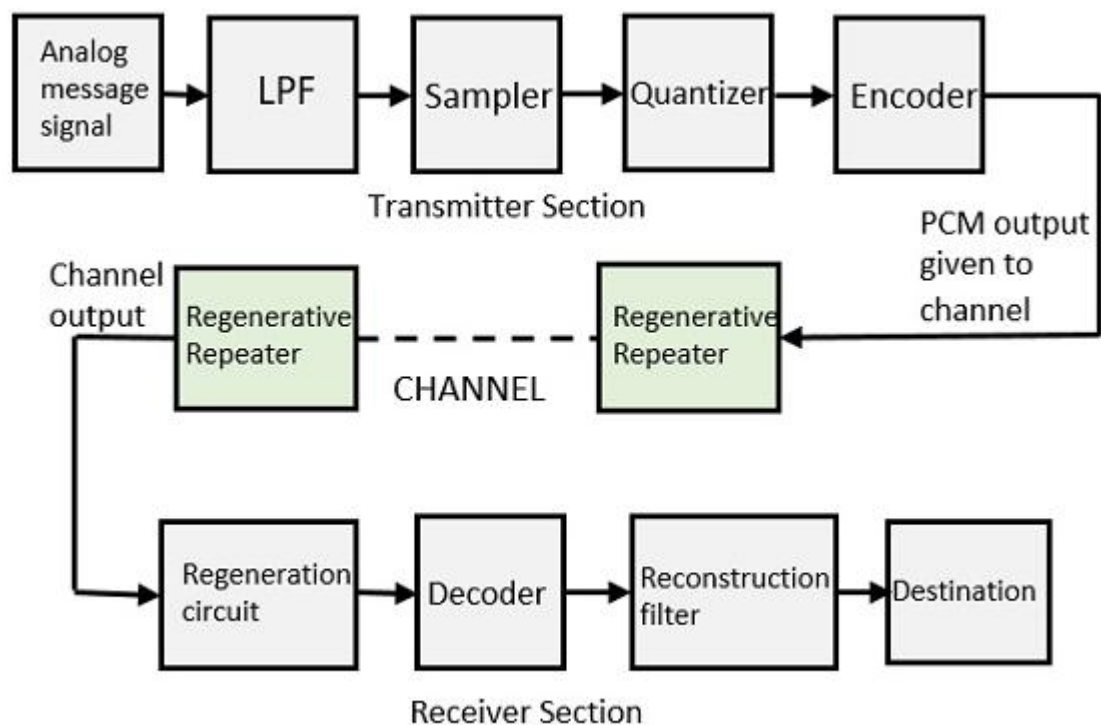
Instead of a pulse train, PCM produces a series of numbers or digits, and hence this process is called as digital. Each one of these digits, though in binary code, represent the approximate amplitude of the signal sample at that instant.

In Pulse Code Modulation, the message signal is represented by a sequence of coded pulses. This message signal is achieved by representing the signal in discrete form in both time and amplitude.

Basic Elements of PCM

The transmitter section of a Pulse Code Modulator circuit consists of Sampling, Quantizing and Encoding, which are performed in the analog-to-digital converter section. The low pass filter prior to sampling prevents aliasing of the message signal.

The basic operations in the receiver section are regeneration of impaired signals, decoding, and reconstruction of the quantized pulse train. Following is the block diagram of PCM which represents the basic elements of both the transmitter and the receiver sections.



1.Low Pass Filter

This filter eliminates the high frequency components present in the input analog signal which is greater than the highest frequency of the message signal, to avoid aliasing of the message signal.

2.Sampler

This is the technique which helps to collect the sample data at instantaneous values of message signal, so as to reconstruct the original signal. The sampling rate must be greater than twice the highest frequency component W of the message signal, in accordance with the sampling theorem.

3.Quantizer

Quantizing is a process of reducing the excessive bits and confining the data. The sampled output when given to Quantizer, reduces the redundant bits and compresses the value.

4.Encoder

The digitization of analog signal is done by the encoder. It designates each quantized level by a binary code. The sampling done here is the sample-and-

hold process. These three sections (LPF, Sampler, and Quantizer) will act as an analog to digital converter. Encoding minimizes the bandwidth used.

5. Regenerative Repeater

This section increases the signal strength. The output of the channel also has one regenerative repeater circuit, to compensate the signal loss and reconstruct the signal, and also to increase its strength.

6. Decoder

The decoder circuit decodes the pulse coded waveform to reproduce the original signal. This circuit acts as the demodulator.

7. Reconstruction Filter

After the digital-to-analog conversion is done by the regenerative circuit and the decoder, a low-pass filter is employed, called as the reconstruction filter to get back the original signal.

Hence, the Pulse Code Modulator circuit digitizes the given analog signal, codes it and samples it, and then transmits it in an analog form. This whole process is repeated in a reverse pattern to obtain the original signal.



Delta Modulation

The type of modulation, where the sampling rate is much higher and in which the stepsize after quantization is of a smaller value Δ , such a modulation is termed as delta modulation.

Features of Delta Modulation

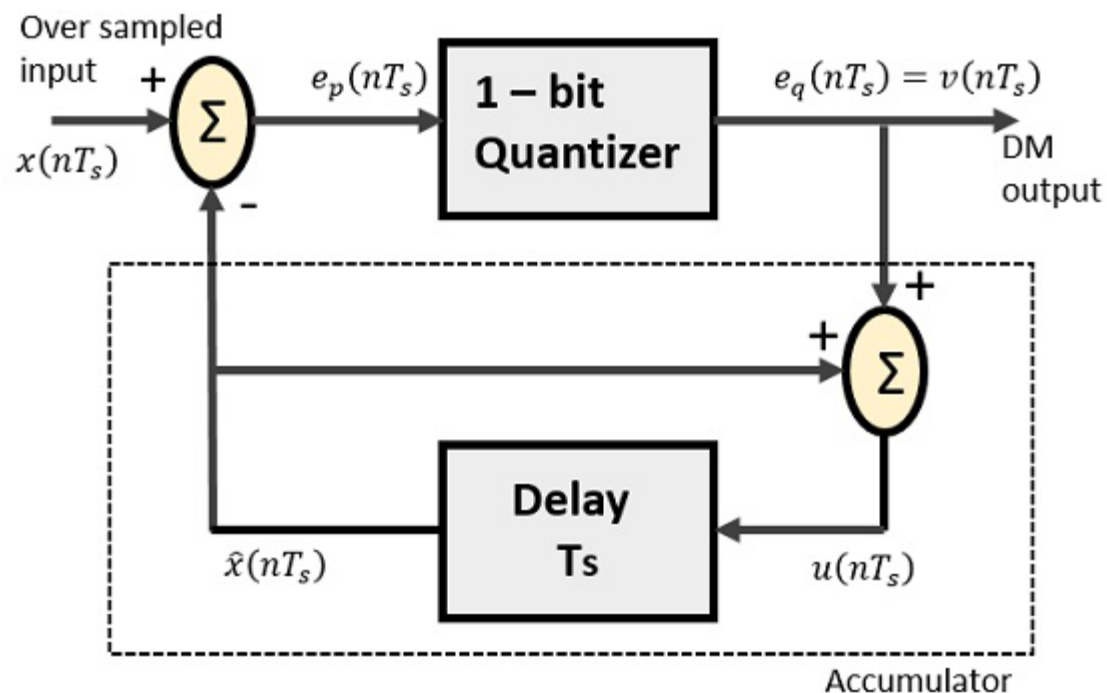
Following are some of the features of delta modulation.

- An over-sampled input is taken to make full use of the signal correlation.
- The quantization design is simple.
- The input sequence is much higher than the Nyquist rate.
- The quality is moderate.
- The design of the modulator and the demodulator is simple.
- The stair-case approximation of output waveform.
- The step-size is very small, i.e., Δ (delta).
- The bit rate can be decided by the user.
- This involves simpler implementation.

Delta Modulation is a simplified form of DPCM technique, also viewed as 1-bit DPCM scheme. As the sampling interval is reduced, the signal correlation will be higher.

Delta Modulator

The Delta Modulator comprises of a 1-bit quantizer and a delay circuit along with two summer circuits. Following is the block diagram of a delta modulator.



The predictor circuit in DPCM is replaced by a simple delay circuit in DM.

From the above diagram, we have the notations as –

- $x(nT_s)$ = over sampled input
- $e_p(nT_s)$ = summer output and quantizer input
- $e_q(nT_s)$ = quantizer output = $v(nT_s)$
- $\hat{x}(nT_s)$ = output of delay circuit
- $u(nT_s)$ = input of delay circuit

Using these notations, now we shall try to figure out the process of delta modulation.

$$e_p(nT_s) = x(nT_s) - \hat{x}(nT_s) \text{-----equation 1}$$

$$= x(nT_s) - u([n-1]T_s)$$

$$= x(nT_s) - [x([n-1]T_s) + v([n-1]T_s)] \text{-----equation 2}$$

Further,

$$v(nT_s) = eq(nT_s) = S \cdot \text{sig}[ep(nT_s)] \text{-----equation 3}$$

$$u(nT_s) = \hat{x}(nT_s) + eq(nT_s)$$

Where,

- $\hat{x}(nT_s)$ = the previous value of the delay circuit
- $eq(nT_s)$ = quantizer output = $v(nT_s)$

Hence,

$$u(nT_s) = u([n-1]T_s) + v(nT_s) \text{-----equation 4}$$

Which means,

The present input of the delay unit

= (The previous output of the delay unit) + (the present quantizer output)

Assuming zero condition of Accumulation,

$$u(nT_s) = S \sum_{j=1}^n \text{sig}[ep(jT_s)]$$

$$\text{Accumulated version of DM output} = \sum_{j=1}^n v(jT_s) \text{-----equation 5}$$

Now, note that

$$\hat{x}(nT_s) = u([n-1]T_s)$$

$$= \sum_{j=1}^{n-1} v(jT_s) \text{-----equation 6}$$

Delay unit output is an Accumulator output lagging by one sample.

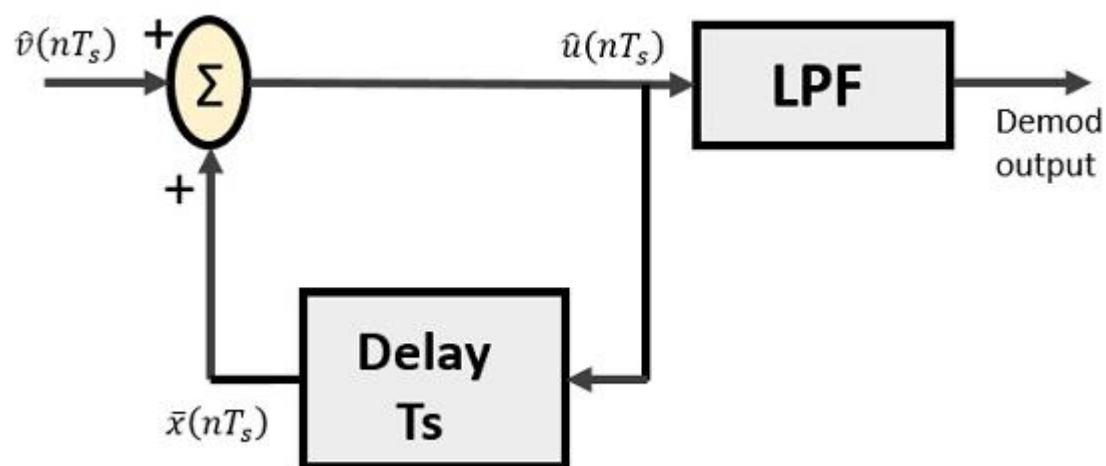
From equations 5 & 6, we get a possible structure for the demodulator.

A Stair-case approximated waveform will be the output of the delta modulator with the step-size as delta (Δ). The output quality of the waveform is moderate.

Delta Demodulator

The delta demodulator comprises of a low pass filter, a summer, and a delay circuit. The predictor circuit is eliminated here and hence no assumed input is given to the demodulator.

Following is the diagram for delta demodulator.



From the above diagram, we have the notations as –

- $\hat{v}(nT_s)$ is the input sample
- $\hat{u}(nT_s)$ is the summer output
- $\hat{x}(nT_s)$ is the delayed output

A binary sequence will be given as an input to the demodulator. The stair-case approximated output is given to the LPF.

Low pass filter is used for many reasons, but the prominent reason is noise elimination for out-of-band signals. The step-size error that may occur at the transmitter is called granular noise, which is eliminated here. If there is no noise present, then the modulator output equals the demodulator input.

Advantages of DM Over DPCM

- 1-bit quantizer
- Very easy design of the modulator and the demodulator

However, there exists some noise in DM.

- Slope Over load distortion (when Δ is small)
- Granular noise (when Δ is large)

Q9. Sampling theorem and Quantization with diagram.

Sampling is the processes of converting continuous-time analog signal $x_a(t)$, into discrete time signal by taking the “samples” at discrete-time intervals. Sampling analog signals makes them discrete in time but still continuous valued.

If done properly (Nyquist theorem is satisfied), sampling does not introduce distortion.

Digital signals are easier to store and have a higher chance of repressing noise. This makes sampling an important step in converting analog signals to digital signals with its primary purpose as representing analog signals in a discrete format.

Important Terminologies of Sampling

Sampled values: – The value of the function at the sampling points

Sampling interval: – The time that separates sampling points (interval b/w samples), T_s . If the signal is slowly varying, then fewer samples per second will be required than if the waveform is rapidly varying.

Sampling Rate (or sampling frequency f_s): The rate at which the signal is sampled, expressed as the number of samples per second. (reciprocal of the sampling interval), $1/T_s = f_s$

Nyquist Sampling Theorem (or Nyquist Criterion):

If the sampling is performed at a proper rate, no info is lost about the original signal and it can be properly reconstructed later on

Statement:

“If a signal is sampled at a rate at least, but not exactly equal to twice the max frequency component of the waveform, then the waveform can be exactly reconstructed from the samples without any distortion”

$$f_s > 2f_{\max}$$

Where,

f_s refers to frequency signal

f_m refers to max frequency

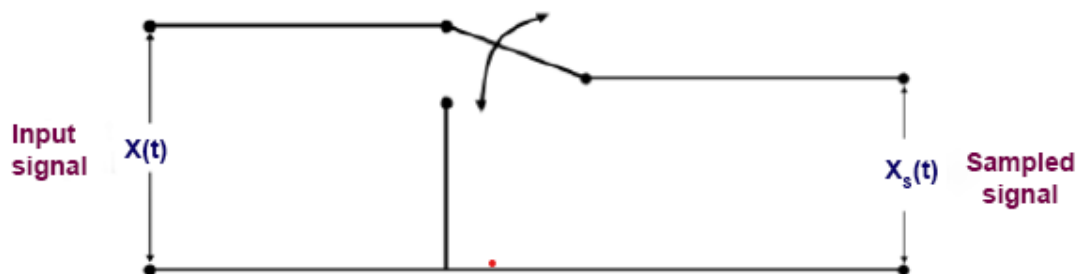
Nyquist Interval:- The Nyquist interval, also known as the Nyquist period, is the time interval between consecutive samples in a digital signal or digital sampling system.

Methods of Sampling

1. Ideal Sampling

Concept: Ideal sampling, also known as impulse or Dirac sampling, is a theoretical notion in which samples of a continuous signal are taken at specific time intervals, often at the Dirac delta function impulse points.

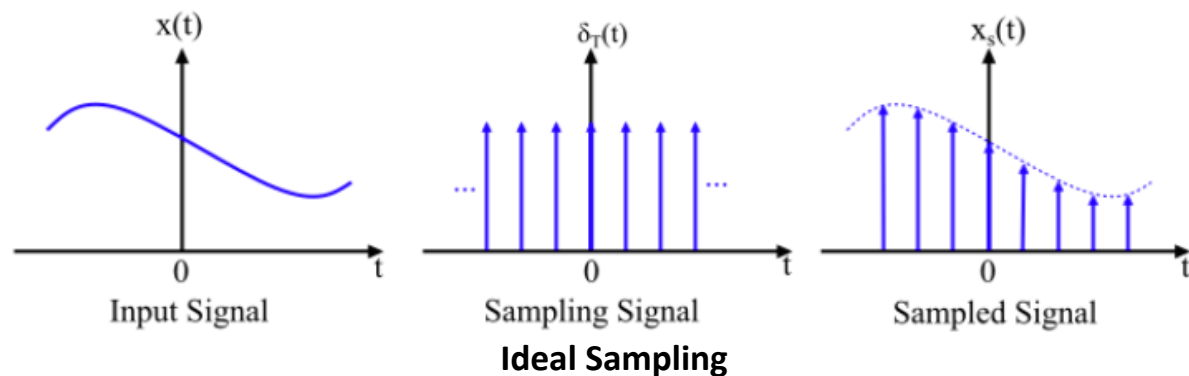
Sampling Process: Each sample in perfect sampling is an impulse or delta function at the sampling instant. The sampled signal can be represented mathematically as the product of the continuous signal and the Dirac delta function.



Sampling Circuit

Reconstruction: The reconstruction of the original signal from ideal samples, we can use interpolation which uses the functions. Ideal sampling is a simple

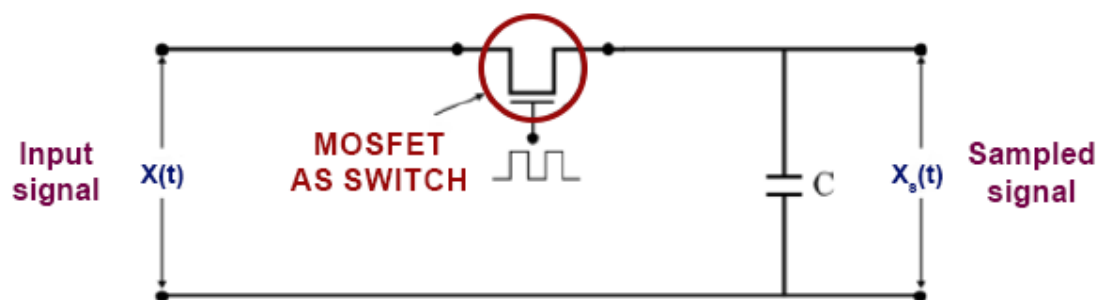
approach to express and analyze sampling theory, however it is not practical due to the requirement for infinite bandwidth.



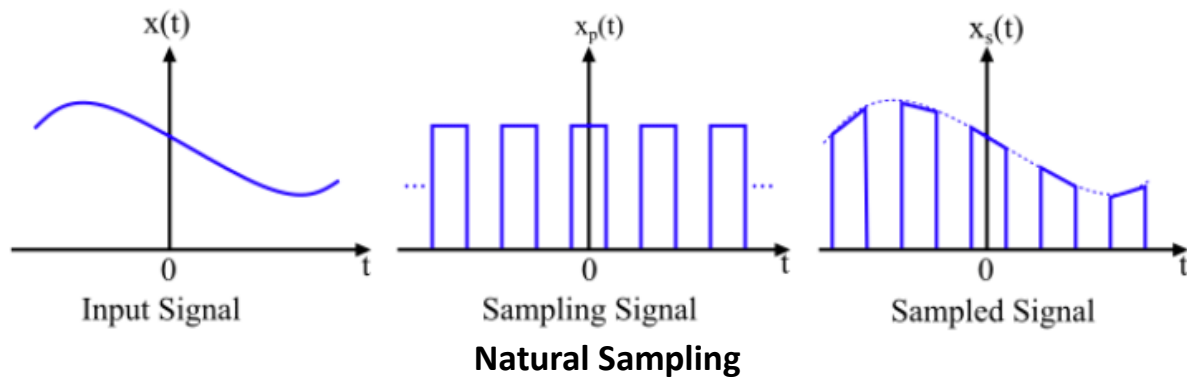
2. Natural Sampling

Concept: Natural sampling, also known as zero-order hold sampling, involves taking discrete interval samples of a continuous signal, similar to uniform sampling. The difference, though, is in how the samples are gathered.

Sampling Process: Each sample is taken in natural sampling by retaining the value of the continuous signal constant for the duration of the sampling period.



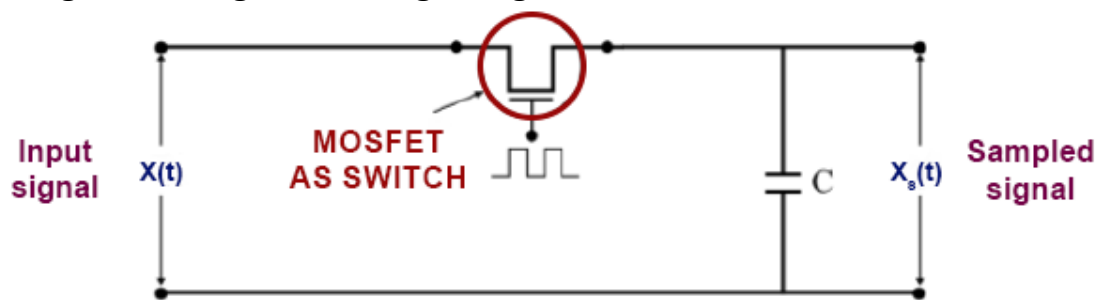
Reconstruction: The reconstruction of the original signal from natural samples, it usually involves connecting the samples with flat line segments. This method simplifies the reconstruction process compared to ideal sampling.



3. Flat-Top Sampling

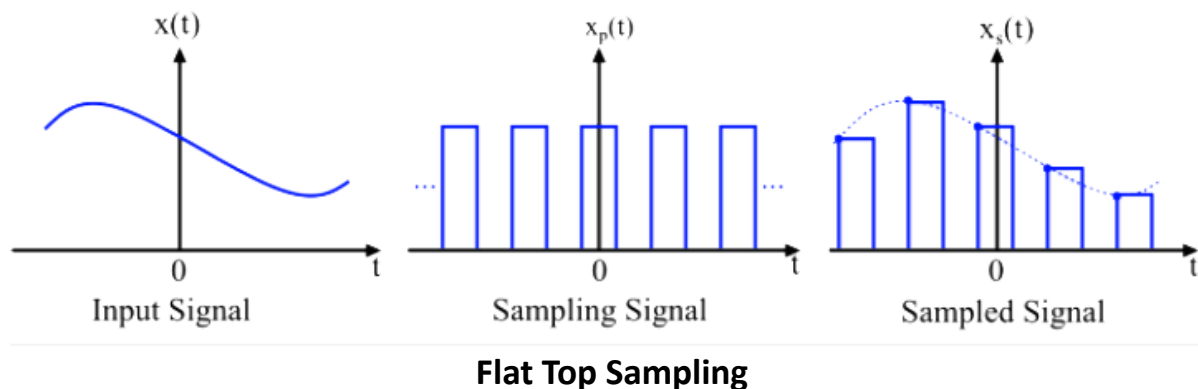
Concept: Flat-top sampling is a type of natural sampling in which each sample is obtained by maintaining the value of the continuous signal constant for a set period of time, resulting in a flat-top waveform.

Sampling Process: Instead of retaining the value for the whole sample interval, flat-top sampling holds it only for a portion of the interval while allowing it to change at the beginning and end.



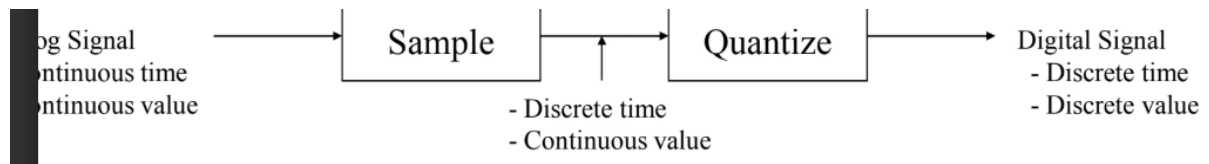
Sampling Circuit

Reconstruction: The reconstruction of the original signal from flat-top samples, we can use interpolation techniques. Flat-top sampling is used in applications where it is desirable to minimize the effects of finite bandwidth and aliasing.



Quantization

Quantization is a non linear transformation which maps elements from a continuous set to a finite set. It is also the second step required by A/D conversion.



Types of Quantization

There are two types of Quantization - Uniform Quantization and Non-uniform Quantization.

The type of quantization in which the quantization levels are uniformly spaced is termed as a **Uniform Quantization**. The type of quantization in which the quantization levels are unequal and mostly the relation between them is logarithmic, is termed as a **Non-uniform Quantization**.

There are two types of uniform quantization. They are Mid-Rise type and Mid-Tread type. The following figures represent the two types of uniform quantization.

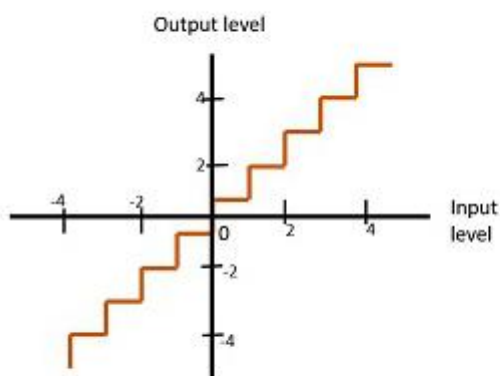


Fig 1 : Mid-Rise type Uniform Quantization

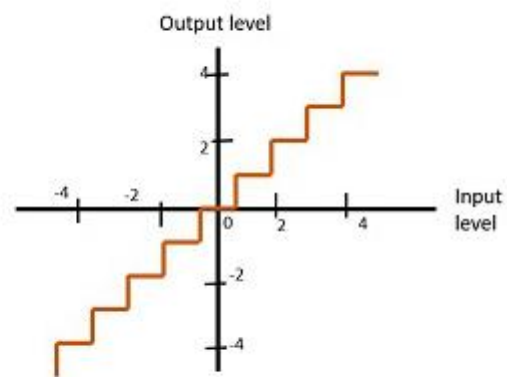


Fig 2 : Mid-Tread type Uniform Quantization

Figure 1 shows the mid-rise type and figure 2 shows the mid-tread type of uniform quantization.

- The **Mid-Rise** type is so called because the origin lies in the middle of a raising part of the stair-case like graph. The quantization levels in this type are even in number.
- The **Mid-tread** type is so called because the origin lies in the middle of a tread of the stair-case like graph. The quantization levels in this type are odd in number.
- Both the mid-rise and mid-tread type of uniform quantizers are symmetric about the origin.

In uniform quantization, the range of the input signal is divided into L equally spaced levels. The step size (Δ) is calculated as:

$$\Delta = \frac{V_{max} - V_{min}}{L}$$

where:

- V_{max} is the maximum value of the input signal.
- V_{min} is the minimum value of the input signal.
- L is the number of quantization levels.

2. Derivation

Let's derive the formula for uniform quantization:

- The range of the input signal is $V_{max} - V_{min}$
- Now, this range is divided into L equal intervals.
- Each interval (or step size) is given by the total range divided by the number of levels, resulting in Δ .

Once the step size is determined, each input value x can be quantized to the nearest level using:

$$Q(x) = \Delta \cdot \left\lfloor \frac{x - V_{min}}{\Delta} + 0.5 \right\rfloor + V_{min}$$

This equation ensures that each input value is mapped to the nearest quantization level.

Non-Uniform quantization on the other hand, uses variable size steps. Instead of equally spaced intervals it uses smaller steps where the signal amplitude is most likely to be and larger step with lower probability. This method is especially one way to overcome the quantization error in case of signals with nonuniform [amplitude](#) distributions, such as speech (such amplitudes depend a lot on each word). Non-uniform quantization can be more difficult to implement but provides better performance for specific applications.

Formula

Non-uniform quantization often uses a companding approach, where the signal is first compressed, uniformly quantized, and then expanded. The compression and expansion functions can vary, but a common method is to use a logarithmic companding function. For example, the μ -law companding used in telephony is given by:

$$y = \text{sgn}(x) \cdot \frac{\ln(1 + \mu |x|)}{\ln(1 + \mu)}$$

where:

- x is the input signal.
- y is the compressed signal.
- μ is the companding parameter.
- $\text{sgn}(x)$ is the sign function, preserving the sign of x .

2. Derivation

The steps for non-uniform quantization using μ law companding are as follows:

1. Compression: Apply the compression function to the input signal x :

$$y = \text{sgn}(x) \cdot \frac{\ln(1 + \mu |x|)}{\ln(1 + \mu)}$$

2. Uniform Quantization: Quantize the compressed signal y using uniform quantization.

3. Expansion: Apply the inverse of the compression function to the quantized value y_q :

$$x_q = \text{sgn}(y_q) \cdot \left(\frac{1}{\mu} ((1 + \mu)^{|y_q|} - 1) \right)$$

This process ensures that the quantization error is minimized in regions where the signal amplitude is more likely to occur, providing better performance for non-uniformly distributed signals.

Q15. Define Aliasing Error with the help of Sampling theorem.

The aliasing effect, also known as aliasing distortion or simply aliasing, is a phenomenon that occurs in signal processing, particularly in digital signal processing (DSP), when a continuous signal is sampled at a frequency that is too low to accurately represent the original signal.

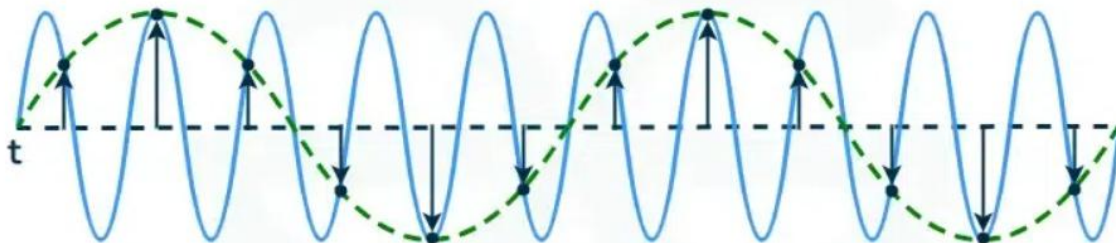
In digital communication, aliasing occurs due to a measurement error in the signal because of an incorrect sampling rate, if the sampling rate is too low aliasing may occur. In simple terms, the [Nyquist rate](#), also known as the Nyquist sampling theorem, is the minimum sampling rate required to accurately sample and reconstruct a signal without distortion. It is defined as twice the maximum frequency present in the signal. Sampling at a rate below the Nyquist rate can lead to aliasing

Points to remember:

- Aliasing is an effect which occurs when the input frequency is half the sampling frequency
- It causes distortion in reconstructed signals
- Anti-aliasing filters are used to prevent aliasing
- Aliasing mainly occurs in digital audio and digital images

Aliasing

Aliasing occurs due to sampling rate being too low with respect to Nyquist Rate



Aliasing occurs due to sampling rate being too low with respect to Nyquist Rate

Cause of Aliasing

Aliasing is mainly caused by poor sampling, if the sampling rate is too low aliasing may occur, aliasing can also occur when signals are not sampled fast enough but mainly aliasing can occur when the sample rate is too low for accurately reconstructing the original signal or when high frequencies in the signal exceed half the sample rate.

Applications of Aliasing

Aliasing mainly has a negative impact on signals, but sometimes it can have positive impact also but it is quite rare. Here are some few applications of Aliasing:

- **To lower frequency of any signal:** Aliasing can be used to convert any high frequency signal into low frequency signal.
- **Medical image processing:** Aliasing is used in medical fields to process signals in their correct form.
- **Facial and text image analysis:** Aliasing has been found useful in analyzing faces and text quite efficiently by using pure sine wave signals.
- **Biometric Recognition:** Aliasing can also recognize biometrics such as fingerprints quite reliably.

Effects of Aliasing

There are a few effects of aliasing and why it should be prevented:

- **Signal Accuracy:** Aliasing makes signal become distorted which can cause unwanted problems in any signal. This can be a major problem in Audio.
- **Reconstruction of Signal:** Due to aliasing, it may become impossible to perfectly reconstruct original signal from its sample because of data loss. Hence aliasing can make reconstruction of signals hard.
- **Signal processing:** Aliasing can make signal processing complicated by producing unwanted noise.
- **Poor signal quality:** Aliasing can negatively impact any signals quality and cause distortion and corruption of the signal.
- **Jagged or blocky appearance:** Aliasing in imagery can also cause jagged edges, pixelated or blocky patterns in the image.

Advantages of Aliasing

Some of the Advantages of Aliasing are :

- **Bandwidth Reduction:** In certain applications where bandwidth is limited, such as in telecommunications or data compression, controlled aliasing techniques can be used to reduce the amount of data that needs to be transmitted or stored.
- **Simulation of Analog Systems:** In certain contexts, aliasing can mimic the behavior of analog systems.
- **Data Analysis:** In [signal processing](#) and data analysis, aliasing can sometimes provide useful insights or simplify analysis.

Disadvantages of Aliasing

- It leads to noise, which can disrupt a signal.
- It leads to distortion and pixelation of any signal.
- It disrupts data signal transmission.
- It can lead to degrade the quality of the signal which can lead to loss of data.
- It can interfere with the accurate detection of signals, leading to missed or false detections.
- It can cause misinterpretation of the signal

Prevention of Aliasing

Aliasing should be prevented to stop distortion in any signal Although aliasing can be prevented just by correctly sampling the signal, there are mainly two methods to prevent aliasing:

Method 1: Using Anti-aliasing Filters

Anti-aliasing filters: These are special filters which can prevent or block any frequencies higher than a specific limit. It simply filters out the unwanted frequencies in any signal.

Advantages of Anti-aliasing Filters

- **Effectively prevents aliasing:** Anti-aliasing filters are a sure-fire or reliable way of prevent aliasing by blocking the unwanted frequencies.
- **Quality Enhancement:** Anti-Aliasing filters can enhance the quality of images by preventing pixelated or distorted parts.
- **Boost Visual Appeal:** Anti-aliasing filters can improve the visual representation of any object

Disadvantages of Anti-Aliasing Filters

- **Time effect:** This refers to the undesirable effects because of the long time responses that occur in digital audio in video, In simple term it means sampling can cause a lot of time to achieve.
- **Phase distortion:** This method cannot prevent aliasing in phase distortions.
- **Unwanted frequency modification:** Sometimes this method will cause the unwanted modification of frequency which can cause another noise to be produced in a signal.

Method 2: Using Oversampling

Oversampling: Oversampling is a technique which is used to measure the signal at a much higher rate than actually needed, which pushes the Nyquist frequency higher. These higher frequencies are then removed without

affecting the desired signal. It is a popular method of reducing Aliasing in digital audio and video.

Advantages of Oversampling

- **Helps preserve the data integrity:** Oversampling does not cause any damage to the original signal or file, which helps promote data integrity and prevents data loss.
- **Reliable:** Oversampling is a reliable way to prevent aliasing in any digital signal.
- **Performance:** Oversampling can make the signal perform better by preventing distortion and also giving it a much higher frequency.

Disadvantages of Oversampling

- **Sampling error:** Oversampling may lead to sampling error since the measured signal is at a higher frequency than the original.
- **Time consuming:** Sampling is a time consuming process overall, Oversampling is even more time consuming because it requires you to measure any signal in its highest frequency
- **Cost and Complexity:** Implementing oversampling techniques may require additional computational power and cause an overall more complex signal to be formed..

Q16. Explain generation and detection of PAM, PPM and PWM with waveforms

Basis for Comparison	PAM	PWM	PPM
Varying parameter	Amplitude	Width	Position
Immunity towards noise	Low	High	High
Signal to noise ratio	Low	Moderate	Comparitively high
Need of synchronization pulse	Not exist	Not exist	Exist
Bandwidth dependency	On pulse width	On rise time of pulse	On rise time of pulse
Transmission power	Variable	Variable	Constant
Bandwidth requirement	Low	High	High
Similarity of implementation	Similar to AM	Similar to FM	Similar to PM
Synchronization between Transmitter and Receiver	Not needed	Not needed	Needed

MOD-5

Q19. Line codes and their desirable properties, PSD of digital data.

A **line code** is the code used for data transmission of a digital signal over a transmission line. This process of coding is chosen so as to avoid overlap and distortion of signal such as inter-symbol interference.

Properties of Line Coding

Following are the properties of line coding –

- As the coding is done to make more bits transmit on a single signal, the bandwidth used is much reduced.
- For a given bandwidth, the power is efficiently used.
- The probability of error is much reduced.
- Error detection is done and the bipolar too has a correction capability.
- Power density is much favorable.
- The timing content is adequate.
- Long strings of **1s** and **0s** is avoided to maintain transparency.

Types of Line Coding

There are 3 types of Line Coding

- Unipolar
- Polar
- Bi-polar

Unipolar Signaling

Unipolar signaling is also called as **On-Off Keying** or simply **OOK**.

The presence of pulse represents a **1** and the absence of pulse represents a **0**.

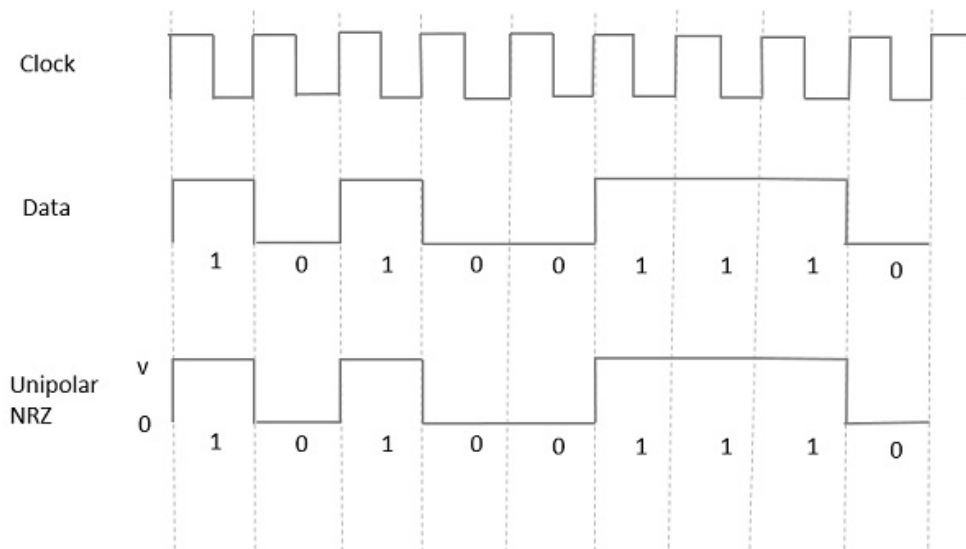
There are two variations in Unipolar signaling –

- Non Return to Zero (NRZ)
- Return to Zero (RZ)

Unipolar Non-Return to Zero (NRZ)

In this type of unipolar signaling, a High in data is represented by a positive pulse called as **Mark**, which has a duration T_0 equal to the symbol bit duration.

A Low in data input has no pulse.



Advantages

The advantages of Unipolar NRZ are –

- It is simple.
- A lesser bandwidth is required.

Disadvantages

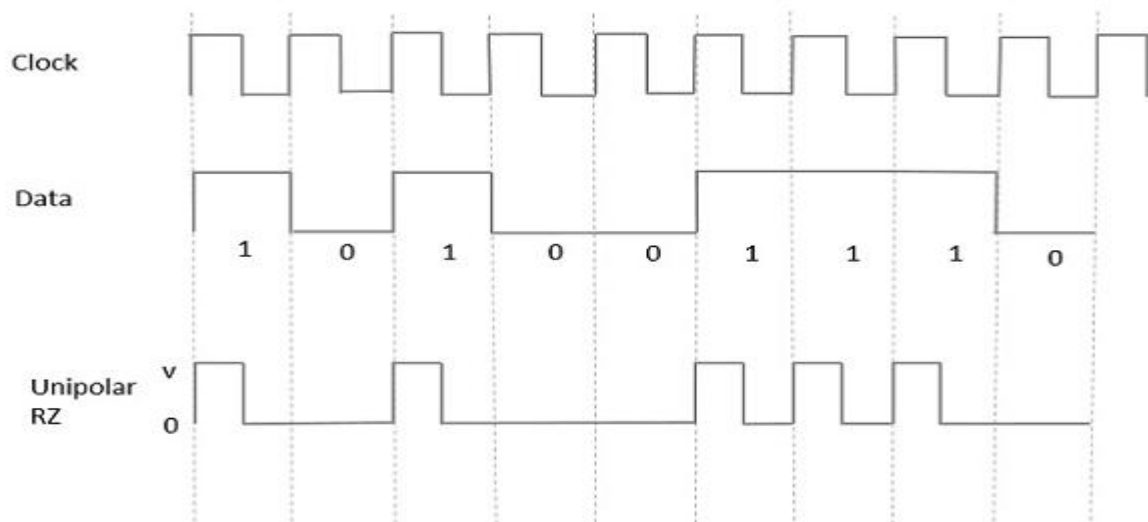
The disadvantages of Unipolar NRZ are –

- No error correction done.
- Presence of low frequency components may cause the signal droop.
- No clock is present.
- Loss of synchronization is likely to occur (especially for long strings of **1s** and **0s**).

Unipolar Return to Zero (RZ)

In this type of unipolar signaling, a High in data, though represented by a **Mark pulse**, its duration T_0 is less than the symbol bit duration. Half of the bit duration remains high but it immediately returns to zero and shows the absence of pulse during the remaining half of the bit duration.

It is clearly understood with the help of the following figure.



Advantages

The advantages of Unipolar RZ are –

- It is simple.
- The spectral line present at the symbol rate can be used as a clock.

Disadvantages

The disadvantages of Unipolar RZ are –

- No error correction.
- Occupies twice the bandwidth as unipolar NRZ.
- The signal droop is caused at the places where signal is non-zero at 0 Hz.

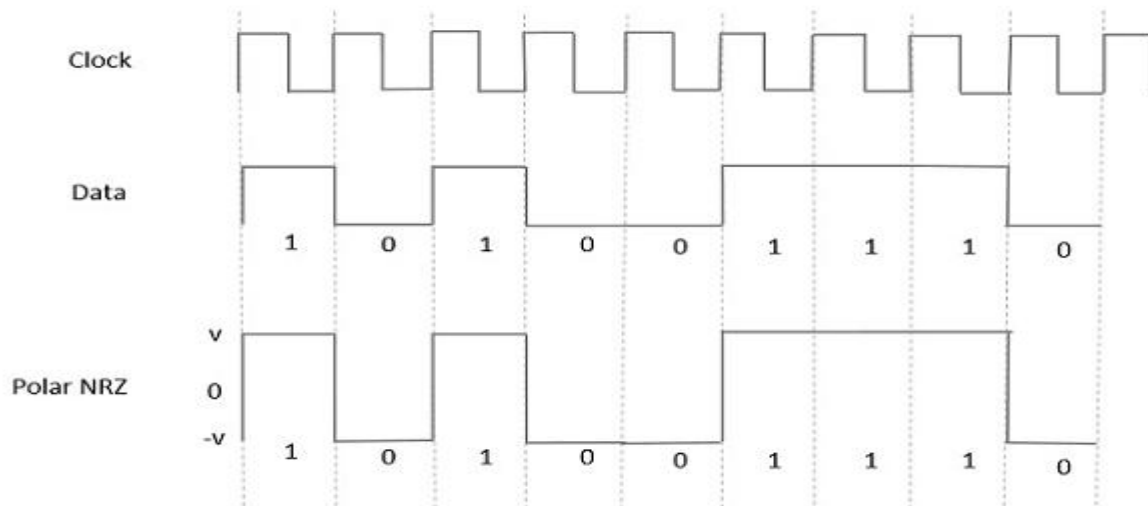
Polar Signaling

There are two methods of Polar Signaling. They are –

- Polar NRZ
- Polar RZ

Polar NRZ

In this type of Polar signaling, a High in data is represented by a positive pulse, while a Low in data is represented by a negative pulse. The following figure depicts this well.



Advantages

The advantages of Polar NRZ are –

- It is simple.
- No low-frequency components are present.

Disadvantages

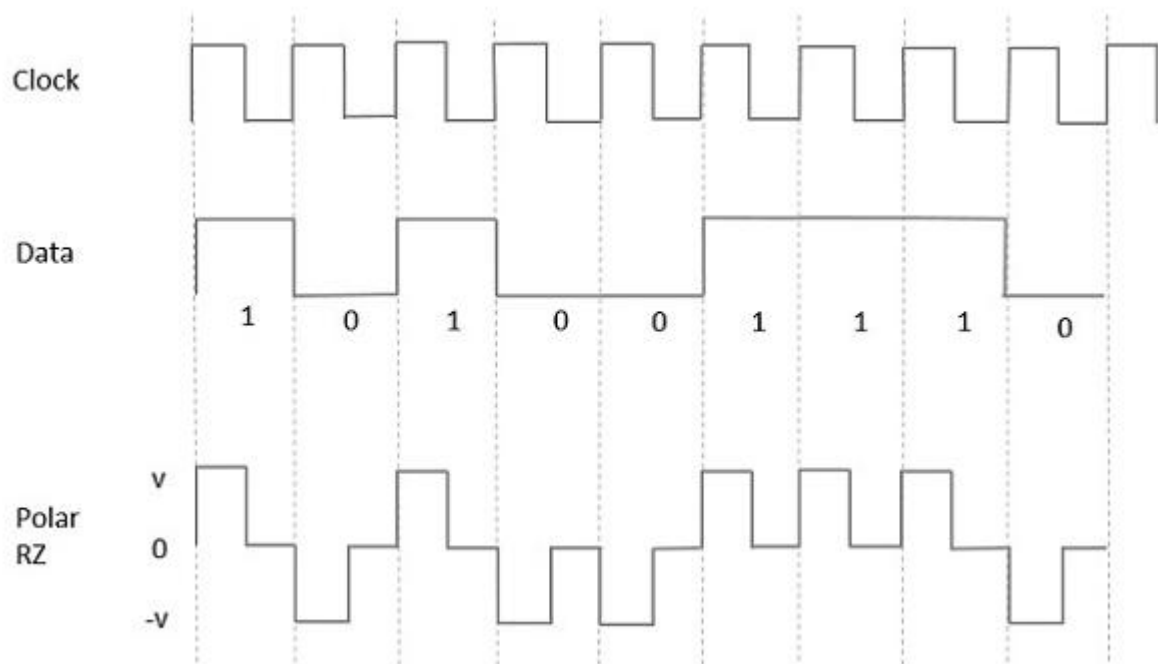
The disadvantages of Polar NRZ are –

- No error correction.
- No clock is present.
- The signal droop is caused at the places where the signal is non-zero at **0 Hz**.

Polar RZ

In this type of Polar signaling, a High in data, though represented by a **Mark pulse**, its duration T_0 is less than the symbol bit duration. Half of the bit duration remains high but it immediately returns to zero and shows the absence of pulse during the remaining half of the bit duration.

However, for a Low input, a negative pulse represents the data, and the zero level remains same for the other half of the bit duration. The following figure depicts this clearly.



Advantages

The advantages of Polar RZ are –

- It is simple.
- No low-frequency components are present.

Disadvantages

The disadvantages of Polar RZ are –

- No error correction.
- No clock is present.
- Occupies twice the bandwidth of Polar NRZ.
- The signal droop is caused at places where the signal is non-zero at **0 Hz**.

Bipolar Signaling

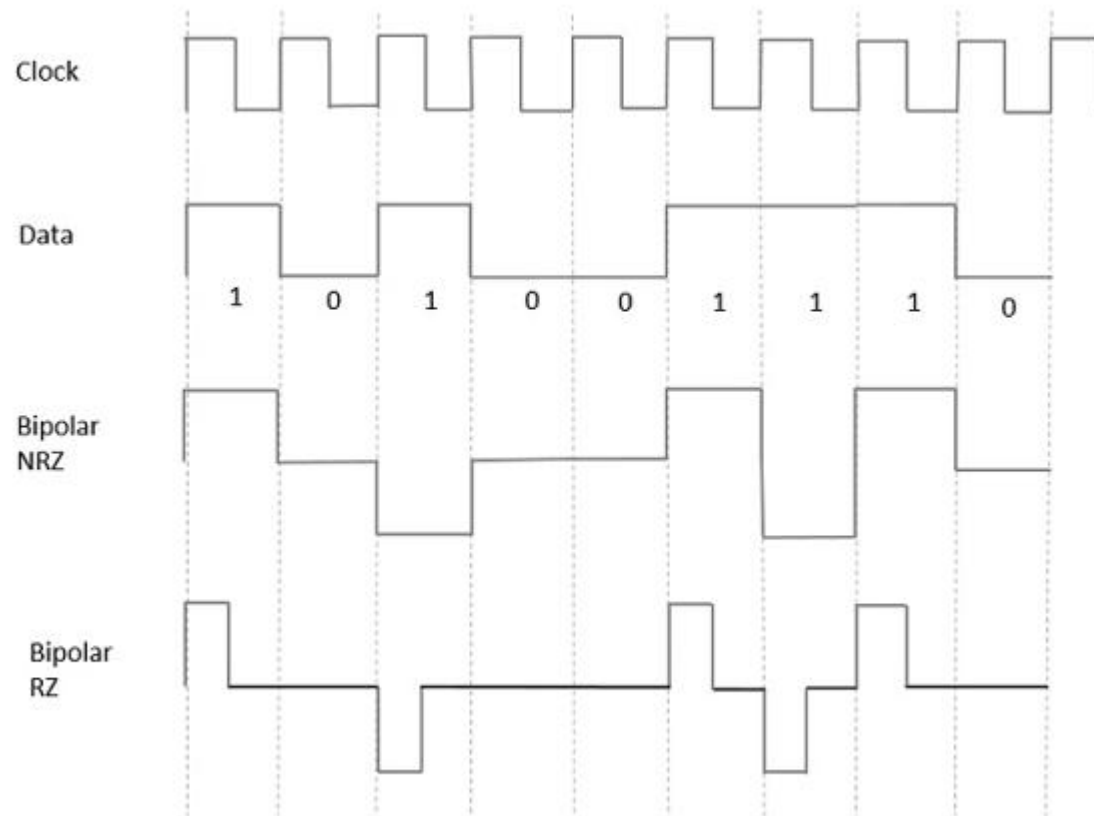
This is an encoding technique which has three voltage levels namely **+**, **-** and **0**. Such a signal is called as **duo-binary signal**.

An example of this type is **Alternate Mark Inversion (AMI)**. For a **1**, the voltage level gets a transition from **+** to **-** or from **-** to **+**, having alternate **1s** to be of equal polarity. A **0** will have a zero voltage level.

Even in this method, we have two types.

- Bipolar NRZ
- Bipolar RZ

From the models so far discussed, we have learnt the difference between NRZ and RZ. It just goes in the same way here too. The following figure clearly depicts this.



The above figure has both the Bipolar NRZ and RZ waveforms. The pulse duration and symbol bit duration are equal in NRZ type, while the pulse duration is half of the symbol bit duration in RZ type.

Advantages

Following are the advantages –

- It is simple.
- No low-frequency components are present.
- Occupies low bandwidth than unipolar and polar NRZ schemes.
- This technique is suitable for transmission over AC coupled lines, as signal drooping doesn't occur here.
- A single error detection capability is present in this.

Disadvantages

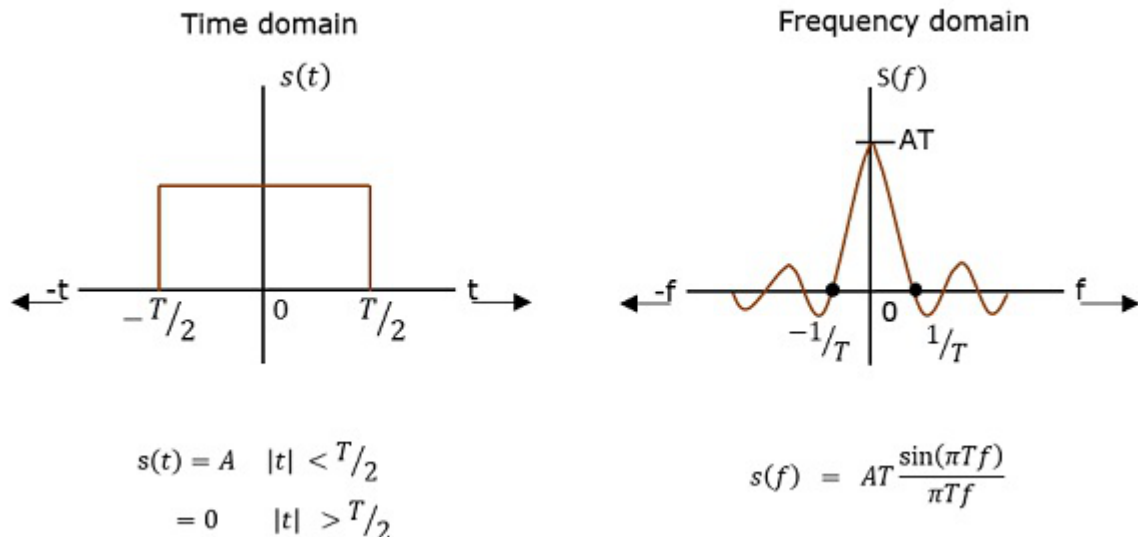
Following are the disadvantages –

- No clock is present.
- Long strings of data causes loss of synchronization.

Power Spectral Density

The function which describes how the power of a signal got distributed at various frequencies, in the frequency domain is called as **Power Spectral Density (PSD)**.

PSD is the Fourier Transform of Auto-Correlation (Similarity between observations). It is in the form of a rectangular pulse.



PSD Derivation

According to the Einstein-Wiener-Khintchine theorem, if the auto correlation function or power spectral density of a random process is known, the other can be found exactly.

Hence, to derive the power spectral density, we shall use the time auto-correlation ($R_x(\tau)$) of a power signal $x(t)$ as shown below.

$$R_x(\tau) = \lim_{T_p \rightarrow \infty} \frac{1}{T_p} \int_{-\frac{T_p}{2}}^{\frac{T_p}{2}} x(t)x(t+\tau)dt$$

Since $x(t)$ consists of impulses, $R_x(\tau)$ can be written as

$$R_x(\tau) = \frac{1}{T} \sum_{n=-\infty}^{\infty} R_n \delta(\tau - nT)$$

Where $R_n = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_k a_k a_{k+n}$

Getting to know that $R_n = R_{-n}$ for real signals, we have

$$S_x(w) = \frac{1}{T} (R_0 + 2 \sum_{n=1}^{\infty} R_n \cos nwT)$$

Since the pulse filter has the spectrum of $(w) \leftrightarrow f(t)$, we have

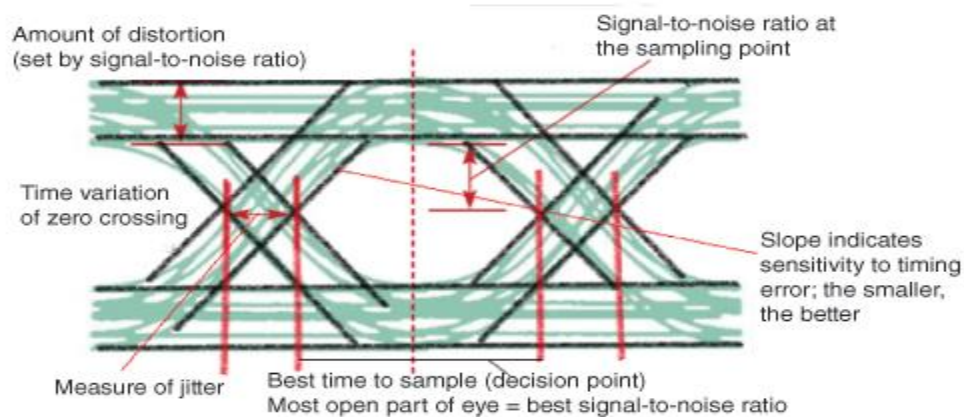
$$\begin{aligned} s_y(w) &= |F(w)|^2 S_x(w) \\ &= \frac{|F(w)|^2}{T} \left(\sum_{n=-\infty}^{\infty} R_n e^{-jnwT_b} \right) \\ &= \frac{|F(w)|^2}{T} (R_0 + 2 \sum_{n=1}^{\infty} R_n \cos nwT) \end{aligned}$$

Hence, we get the equation for Power Spectral Density. Using this, we can find the PSD of various line codes.

Q13. Explain and Analyze EYE diagram with diagram.

An eye diagram is a common indicator of the quality of signals in high-speed digital transmissions. An [oscilloscope](#) generates an eye diagram by overlaying sweeps of different segments of a long data stream driven by a master clock. The triggering edge may be positive or negative, but the displayed pulse that appears after a delay period may go either way; there is no way of knowing beforehand the value of an arbitrary bit. Therefore, when many such transitions have been overlaid, positive and negative pulses are superimposed on each other. Overlaying many bits produces an eye diagram, so called because the resulting image looks like the opening of an eye.

An effective way to study the effects of ISI is the Eye Pattern. The name Eye Pattern was given from its resemblance to the human eye for binary waves. The interior region of the eye pattern is called the eye opening. The following figure shows the image of an eye-pattern.



Jitter is the short-term variation of the instant of digital signal, from its ideal position, which may lead to data errors. Jitter occurs when a rising or falling edges occur at times that differ from the ideal time. Some edges occur early, some occur late. In a digital circuit, all signals are transmitted in reference to clock signals.

When the effect of ISI increases, traces from the upper portion to the lower portion of the eye opening increases and the eye gets completely closed, if ISI is very high.

An eye pattern provides the following information about a particular system.

- Actual eye patterns are used to estimate the bit error rate and the signal-to-noise ratio.
- The width of the eye opening defines the time interval over which the received wave can be sampled without error from ISI.
- The instant of time when the eye opening is wide, will be the preferred time for sampling.
- The rate of the closure of the eye, according to the sampling time, determines how sensitive the system is to the timing error.
- The height of the eye opening, at a specified sampling time, defines the margin over noise.

Hence, the interpretation of eye pattern is an important consideration.

Eye diagrams provide instant visual data that engineers can use to check the signal integrity of a design and uncover problems early in the design process.

Used in conjunction with other measurements such as bit-error rate, an eye diagram can help a designer predict performance and identify possible sources of problems.

Q3. Inter symbol interference with its Application and how to avoid it.

There is another important distortion which is most likely to occur, called as **Inter-Symbol Interference (ISI)**.

This is a form of distortion of a signal, in which one or more symbols interfere with subsequent signals, causing noise or delivering a poor output.

Causes of ISI

The main causes of ISI are –

- Multi-path Propagation
- Non-linear frequency in channels

The ISI is unwanted and should be completely eliminated to get a clean output.

The causes of ISI should also be resolved in order to lessen its effect.

To view ISI in a mathematical form present in the receiver output, we can consider the receiver output.

The receiving filter output $y(t)$ is sampled at time $t_i = iT_b$ (with i taking on integer values), yielding –

$$\begin{aligned} y(t_i) &= \mu \sum_{k=-\infty}^{\infty} a_k p(iT_b - kT_b) \\ &= \mu a_i + \mu \sum_{\substack{k=-\infty \\ k \neq i}}^{\infty} a_k p(iT_b - kT_b) \end{aligned}$$

In the above equation, the first term μa_i is produced by the i^{th} transmitted bit. The second term represents the residual effect of all other transmitted bits on the decoding of the i^{th} bit. This residual effect is called as **Inter Symbol Interference**.

In the absence of ISI, the output will be –

$$y(t_i) = \mu a_i$$

This equation shows that the i^{th} bit transmitted is correctly reproduced.

However, the presence of ISI introduces bit errors and distortions in the output.

While designing the transmitter or a receiver, it is important that you minimize the effects of ISI, so as to receive the output with the least possible error rate.

So far, we've discussed that ISI is an unwanted phenomenon and degrades the signal. But the same ISI if used in a controlled manner, is possible to achieve a bit rate of **2W** bits per second in a channel of bandwidth **W** Hertz. Such a scheme is called as **Correlative Coding** or **Partial response signaling schemes**.

Since the amount of ISI is known, it is easy to design the receiver according to the requirement so as to avoid the effect of ISI on the signal. The basic idea of

correlative coding is achieved by considering an example of **Duo-binary Signaling**.

Duo-binary Signaling

The name duo-binary means doubling the binary systems transmission capability. To understand this, let us consider a binary input sequence $\{a_k\}$ consisting of uncorrelated binary digits each having a duration T_a seconds. In this, the signal **1** is represented by a **+1** volt and the symbol **0** by a **-1** volt.

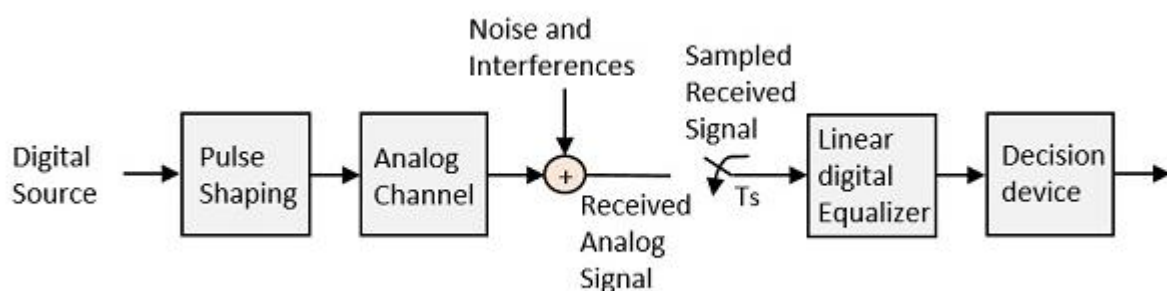
Therefore, the duo-binary coder output c_k is given as the sum of present binary digit a_k and the previous value a_{k-1} as shown in the following equation.

$$c_k = a_k + a_{k-1}$$

The above equation states that the input sequence of uncorrelated binary sequence $\{a_k\}$ is changed into a sequence of correlated three level pulses $\{c_k\}$. This correlation between the pulses may be understood as introducing ISI in the transmitted signal in an artificial manner.

For reliable communication to be established, we need to have a quality output. The transmission losses of the channel and other factors affecting the quality of the signal, have to be treated. The most occurring loss, as we have discussed, is the ISI.

To make the signal free from ISI, and to ensure a maximum signal to noise ratio, we need to implement a method called **Equalization**. The following figure shows an equalizer in the receiver portion of the communication system.



The noise and interferences which are denoted in the figure, are likely to occur, during transmission. The regenerative repeater has an equalizer circuit, which compensates the transmission losses by shaping the circuit. The Equalizer is feasible to get implemented.

Error Probability and Figure-of-merit

The rate at which data can be communicated is called the **data rate**. The rate at which error occurs in the bits, while transmitting data is called the **Bit Error Rate (BER)**.

The probability of the occurrence of BER is the **Error Probability**. The increase in Signal to Noise Ratio (SNR) decreases the BER, hence the Error Probability also gets decreased.

In an Analog receiver, the **figure of merit** at the detection process can be termed as the ratio of output SNR to the input SNR. A greater value of figure-of-merit will be an advantage.

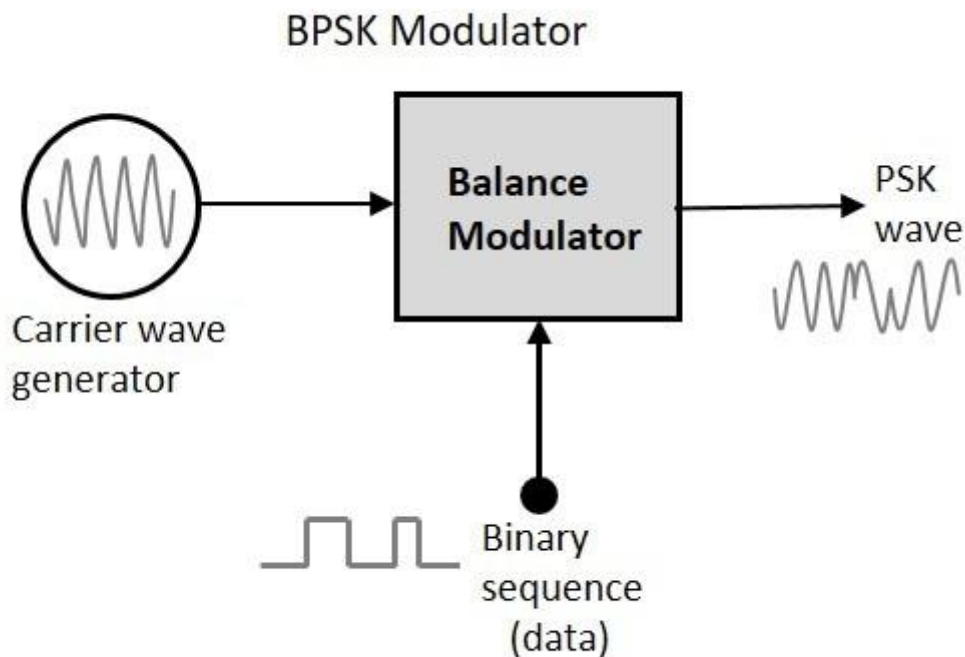
MOD-6

Q7. Explain BFSK generation and reception with block diagram and equations.

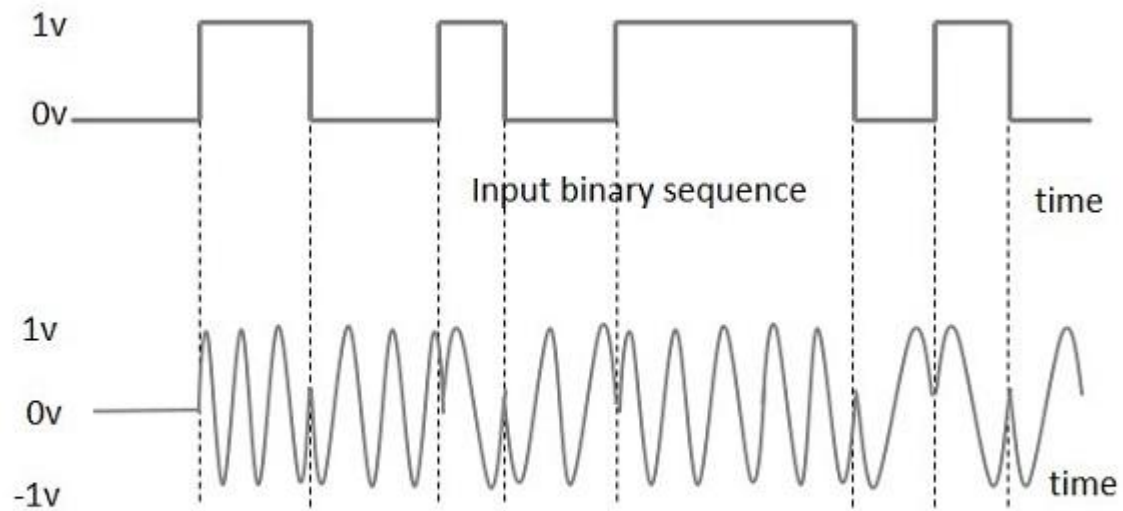
Binary Phase Shift Keying (BPSK) is a widely used modulation technique in digital communication systems for transmitting binary data over a communication channel. It is known for its simplicity and effectiveness, making it a popular choice in applications where the communication channel is susceptible to noise and interference.

In BPSK the carrier signal is modified by altering its phase by 180 degrees, for each symbol. A phase shift of 180 degrees denotes a binary 0 while no phase shift represents a binary 1. The BPSKs [modulation](#) process is straightforward and efficient making it suitable for scenarios where the communication channel suffers from noise and interference. BPSK is basically a Double Side Band Suppressed Carrier (DSBSC) modulation scheme, for message being the digital information.

The block diagram of Binary Phase Shift Keying consists of the balance modulator which has the carrier sine wave as one input and the binary sequence as the other input. Following is the diagrammatic representation.



The modulation of BPSK is done using a balance modulator, which multiplies the two signals applied at the input. For a zero binary input, the phase will be **0°** and for a high input, the phase reversal is of **180°**. Following is the diagrammatic representation of BPSK Modulated output wave along with its given input.



BPSK Modulated output wave

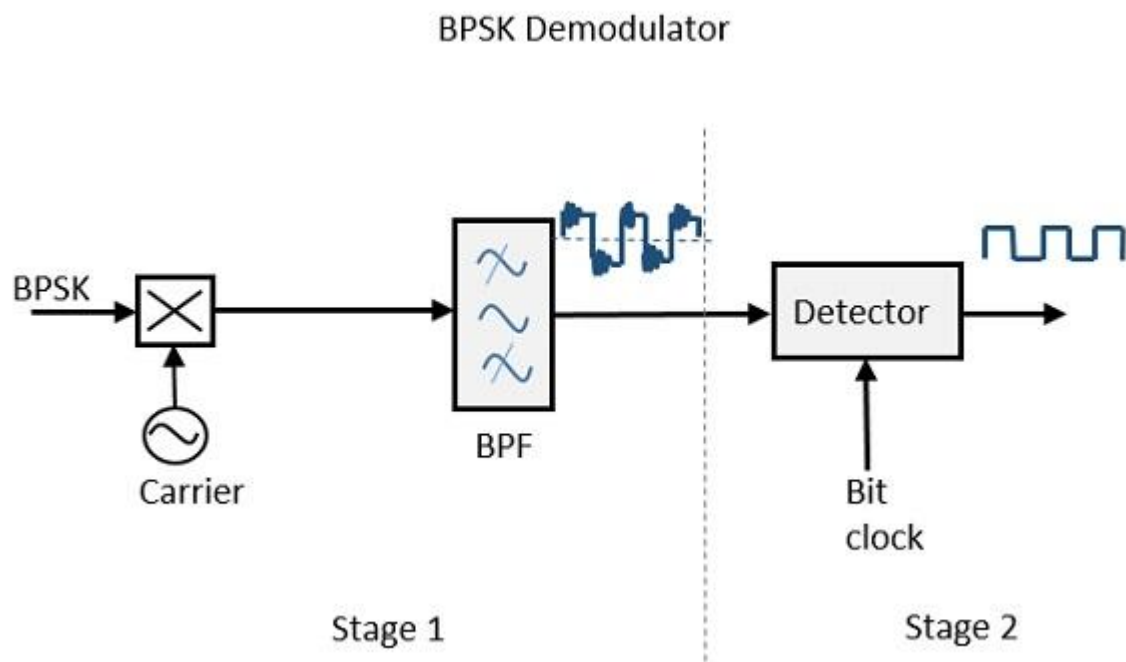
The output sine wave of the modulator will be the direct input carrier or the inverted (180° phase shifted) input carrier, which is a function of the data signal.

Finally, we have

$$s_1(t) = \sqrt{E_b} \phi_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos 2\pi f_c t$$

$$s_0(t) = -\sqrt{E_b} \phi_1(t) = -\sqrt{\frac{2E_b}{T_b}} \cos 2\pi f_c t$$

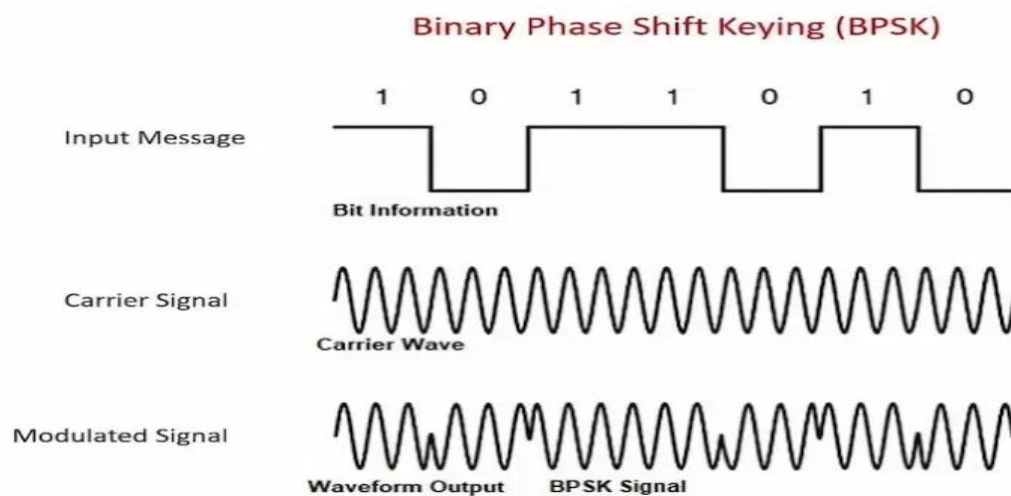
The block diagram of BPSK demodulator consists of a mixer with local oscillator circuit, a bandpass filter, a two-input detector circuit. The diagram is as follows.



The base band signal which is band limited is obtained and this signal is used to regenerate the binary message bit stream.

In the next stage of demodulation, the bit clock rate is needed at the detector circuit to produce the original binary message signal.

Complete Waveform Simulation for BPSK Modulation and Demodulation



Advantages of BPSK

There are multiple advantages of Binary Phase Shift Keying process including all signals form, such as:

- **Simplicity:** BPSK, which stands for Binary Phase Shift Keying is a modulation scheme. It simplifies implementation, in hardware and software by utilizing two phase states; 0 degrees and 180 degrees.
- **Effective Operation with Reliability:** It has ability to operate effectively in the presence of noise or interference from signals ensuring reliable performance.
- **Less Power Consumption:** BPSK consumes [power](#) compared to alternative methods making it advantageous for [battery](#) powered devices.
- **Easy Detection:** Receivers find it easy to comprehend BPSK accurately determining the frequency and phase of the transmitted signal.
- **Compatible:** BPSK serves as a building block for complex modulation.

Disadvantages of BPSK

Some of the disadvantages of Binary Phase Shift Keying process includes:

- **Low Data Sending Rate:** However when it comes to data transmission speed, BPSK has limitations as it can only send one piece of data at a time.
- **Less Efficient:** It inefficiently uses the signal space, same like using a whole road for just one small car.
- **Can be Tricky:** In situations such as communication, signal bouncing can pose challenges, for BPSK as it weakens the signal strength and leads to potential issues.
- **Not for Huge Data:** If you need to send a lot of data fast then BPSK might not be the best choice.

Q11. Differentiate between BASK, BPSK and BFSK.

Feature	BASK	BPSK	BFSK
Parameter varied	Amplitude	Phase	Frequency
Power efficiency	Low	High	Moderate
Bandwidth efficiency	High	Moderate	Low

Feature	BASK	BPSK	BFSK
Error performance	Poor	Excellent	Good
Complexity	Low	Moderate	Moderate
Noise Performance	Poor (affected by amplitude noise)	Best (very robust to noise)	Better than BASK, but less than BPSK
Applications	Low-speed modems, optical communication	Satellite communication, Wi-Fi, RFID	Radio control systems, modems, telemetry